

BIG BAY, MI, USA

WMO: 997988

Lat: 46.8269N

Lon: 87.7266W

Elev: 607

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -0.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 26.6	(k) 21.5	(l) 24.3	(m) 20.1	(n) 5.2	(o) 230	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.9	(c) 84.4	(d) N/A	(e) 81.0	(f) N/A	(g) 77.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.9	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.2	(b) 18.7	(c) 16.0	(d) DB	(e) -12.3	(f) 90.2	(g) 8.8	(h) 2.0	(i) -18.6	(j) 91.7	(k) -23.8	(l) 92.8	(m) -28.7	(n) 93.9	(o) -35.1	(p) 95.3	(q) N/A
(a) 23.2	(b) 18.7	(c) 16.0	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	18.0	17.9	27.7	38.2	48.5	57.6	66.8	65.9	59.3	46.8	35.6	24.6
	DBStd	19.06	10.33	10.23	10.92	7.57	8.32	7.11	6.32	5.80	7.55	7.79	9.11	8.44
	HDD50	4487	993	899	697	364	129	13	0	0	8	157	440	787
	HDD65	8508	1458	1319	1157	803	516	240	56	56	201	567	883	1252
	CDD50	1709	0	0	5	11	83	242	522	493	288	58	7	0
	CDD65	255	0	0	0	0	5	19	112	85	31	3	0	0
	CDH74	1689	0	0	9	1	128	181	661	496	190	23	0	0
	CDH80	381	0	0	0	0	52	50	127	106	43	3	0	0
(14) Wind	WSAvg	5.6	6.7	6.6	6.1	5.9	5.1	4.0	4.7	4.6	5.4	5.9	6.2	6.2
(15) Precipitation	PrecAvg	30.5	1.6	1.3	2.0	2.4	3.0	3.2	3.5	2.4	3.6	3.3	2.3	1.9
	PrecMax	39.8	4.6	3.5	3.9	4.9	6.3	5.4	6.1	5.6	7.3	5.6	4.1	4.2
	PrecMin	22.6	0.6	0.1	0.1	0.7	1.2	0.8	1.5	0.8	1.4	1.5	1.1	0.2
	PrecStd	4.3	0.9	0.7	1.0	1.2	1.3	1.2	1.2	1.1	1.6	1.3	0.8	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	46.1	71.4	70.9	87.8	86.3	87.5	88.6	85.8	77.2	63.9	48.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	39.8	39.9	55.1	61.9	76.4	79.8	83.5	83.0	79.0	69.9	57.2	42.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	36.1	35.6	49.5	55.8	68.4	74.1	80.4	79.3	75.1	64.3	52.7	38.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.9	32.2	43.3	50.7	62.6	69.6	77.5	76.4	71.5	59.1	49.0	36.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	16.2	18.4	15.6	17.9	17.7	16.9	16.3	16.6	13.6	11.8	10.6
		MCDBR	15.8	20.0	24.4	25.8	27.1	24.1	20.2	19.8	20.8	22.5	19.1	14.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.269	0.286	0.314	0.343	0.379	0.382	0.389	0.372	0.363	0.332	0.301	0.267
	taud		2.440	2.377	2.372	2.379	2.362	2.388	2.403	2.448	2.468	2.518	2.521	2.491
	Ebn at Noon		266	283	290	290	281	280	276	277	270	263	252	255
	Edh at Noon		21	28	32	36	37	37	36	33	30	24	20	18
(44) All-Sky Solar Radiation	RadAvg		324	597	1061	1438	1743	1895	1930	1643	1201	707	364	245
	RadStd		26	75	109	155	123	140	97	107	74	85	50	25

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page